Rugved Santosh Kulkarni

Second-Year Computer Science and Engineering Student
Department of Integrated B.Tech - 2025

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Degree	Institute	Year	CPI/%
Integrated B.Tech CSE	Sanjivani University, Kopargaon	2025-2026	9.05 CGPA
Matriculation (SSC)	Seva Niketan Convent School, Kopargaon	2024	85.40%

Work Experience

SmartLab , Sanjivani University

Apr 2026 – Present

Role	• Student Researcher – AI/ML Research, Computer Vision, Embedded Systems
Focus Areas	• Investigating advanced AI algorithms, lightweight object detection models, and intelligent embedded computing platforms. • Working on artificial intelligence, computer vision, machine learning, and embedded system applications for real-world problem solving.
Key Activities	• Developing research prototypes, evaluating model performance, and supporting AI research through experimentation and technical documentation.

Relevant Projects

AI-Powered SEM Wafer Defect Inspection & Navigation Recovery — *GitHub* — *Live Demo*

Aug 2026

Objective	• Develop an AI-based SEM wafer inspection pipeline for automated semiconductor defect segmentation and visual analysis.
Approach	• Audited and prepared a COCO-annotated SEM dataset, converting polygon annotations into pixel-level masks. • Trained U-Net++,DeepLabV3+,SegFormer,HRNet,and AttentionU-Net, integrating the best-performing model into DefectVisionAI.
Impact	• Defect analysis for pixel-level localization,identification,confidence estimation, matching, positioning, and navigation-drift correction.

Design and Development of AI-Enhanced Hyperspectral Eye Disease Detection System — *GitHub*

Aug 2026

Objective	• Developed an AI-assisted eye disease screening system integrating fundus imaging, hyperspectral imaging, and ROI-based spectral analysis.
Approach	• Implemented a multimodal workflow combining deep-learning-based disease screening with user-selected ROI spectral analysis across the 550–620 nm wavelength range. • Evaluated deep learning architectures including Swin Transformer, EfficientNetV2-S, ResNeXt50, FNet, and Perceiver IO for retinal image classification. • Developed interactive visualization for disease prediction, confidence scores, detected regions, spectral signatures, and intensity–wavelength graphs.
Impact	• Built an integrated research prototype enabling spatial + spectral retinal analysis and AI-assisted screening in a unified platform.

Real-Time AI-Based Esophageal Cancer Detection Using Lightweight Object Detection

2025

Objective	• Developed an AI-enabled clinical diagnostic framework for real-time detection and localization of esophageal cancer lesions from endoscopic images and videos.
Approach	• Trained and compared YOLOv8, YOLOv11, Faster R-CNN, SSD, and RT-DETR using Precision, Recall, F1-score, Accuracy, mAP, FPS, CPU, GPU, and RAM. • Developed a PyQt5-based clinical application supporting live camera and pre-recorded video detection, lesion localization, confidence visualization, and patient report generation.
Impact	• Developed an end-to-end AI-assisted prototype focused on real-time lesion detection and the accuracy–latency trade-off for practical clinical deployment.

AI-Powered Onion Quality Inspection & Grading System — *GitHub*

sept 2026

Objective	• Develop an automated AI-based system for onion quality inspection, classification, sizing, counting, and grading.
Approach	• Implemented a PyQt5-based inspection application integrating ConvNeXt Tiny, EfficientNetV2-B0, DenseNet121, ResNet50V2, and Xception models for four-class onion-condition classification. Intel RealSense depth sensing, OpenCV-based object tracking, physical size estimation, Grad-CAM explainability, and Arduino-based conveyor/servo control.
Impact	• Enables real-time inspection with persistent object tracking, batch counting, disease/quality classification, depth-assisted sizing, grading information, and visual explanations for operator review.

Technical Skills

Programming Languages & AI/ML

Languages	• C++, Python, C , HTML/CSS
AI/ML Libraries	• Pandas, NumPy, OpenCV, TensorFlow, Scikit-learn
Embedded & Hardware	• Raspberry Pi, Embedded Systems, IoT Prototyping
Tools & Platforms	• GitHub, Android Studio, VS Code,, Google Colab, Jupyter,
Core Competencies	• Object Detection, Data Analysis, Web Development, Computer Vision, Data Analytics

Certifications & Achievements

Competitions & Hackathons

Workshops	• Google Cloud Arcade Workshop – Completed hands-on cloud computing and Google Cloud platform fundamentals.– *Link* • IIT Bombay DS Workshop – Participated in a workshop on Data Structures and Algorithms.– *Link* • ESP32 Workshop – Gained practical experience with ESP32 microcontrollers and IoT application development..– *Link*
Participant	• DIPEX Hackathon 2026, Sanjivani University (14th Feb 2026) – "Brain-Controlled Robotic Car Using Bio-Signals," a hardware-based assistive system designed to enable hands-free mobility for physically disabled individuals. – *Link* • Google Student Ambassador Program – Game Night Edition – *Link* • Smart India Hackathon (SIH) 2025 – Participated in a national-level innovation hackathon, developing a real-world technology solution through collaborative problem-solving. – *Link*